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The Linac Laser Notcher for the Fermilab Booster

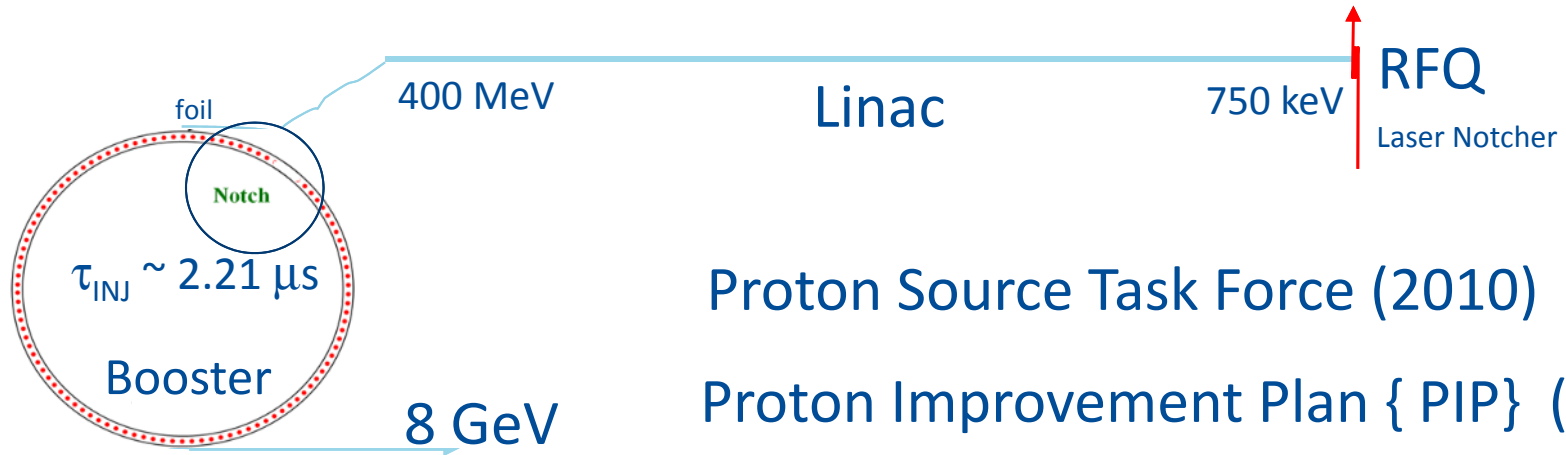
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LINAC 2016 MSU, Lansing MI
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- KV Reddy & Sreenivas Patel of [PriTel](#)
- Don Sipes & Jason Tafoya of [Optical Engines](#)
- Jay Doster & Dennis Lockwood [Grumman](#)

A Little Background (The Proton Source)



Proton Source Task Force (2010)

Proton Improvement Plan { PIP } (2012)

$\mathcal{F}_{\text{RF}}$ (injection) 38 MHz

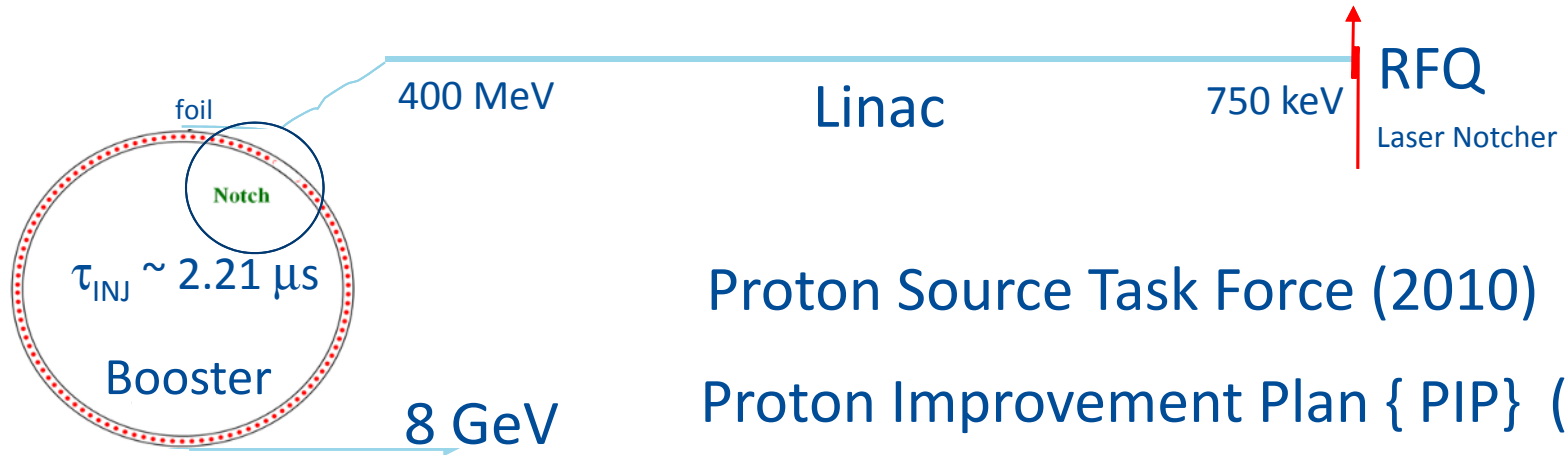
❑ Notch created in Booster @ 400 MeV -> 1/3 total power loss

➤ Move this process out of the Booster tunnel

➤ Create Notches in Linac pulse



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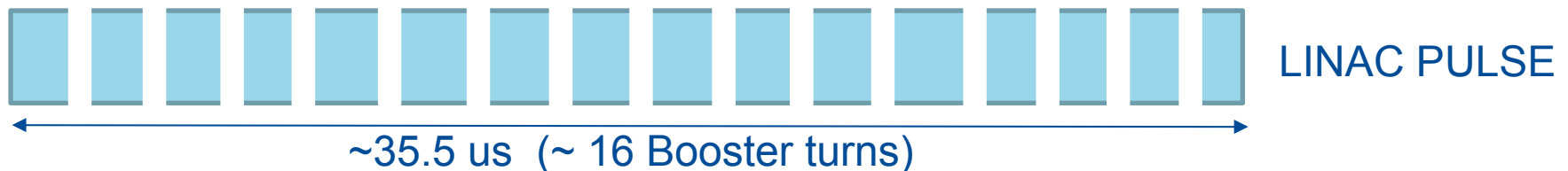
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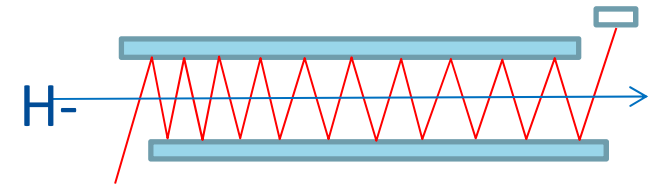
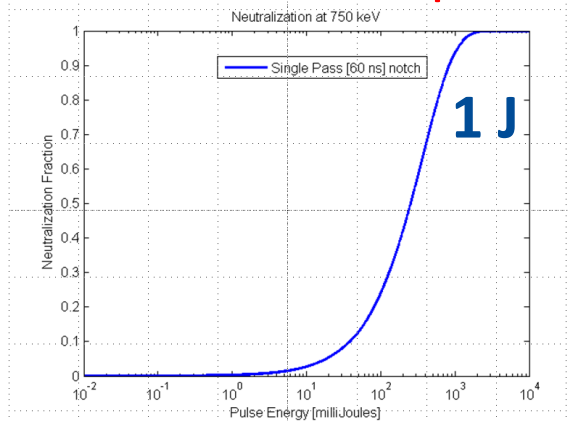
➤ Create Notches in Linac pulse



Technique: Photoneutralization

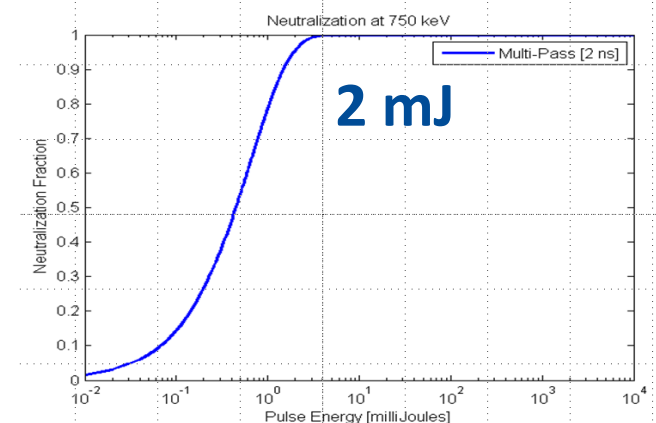
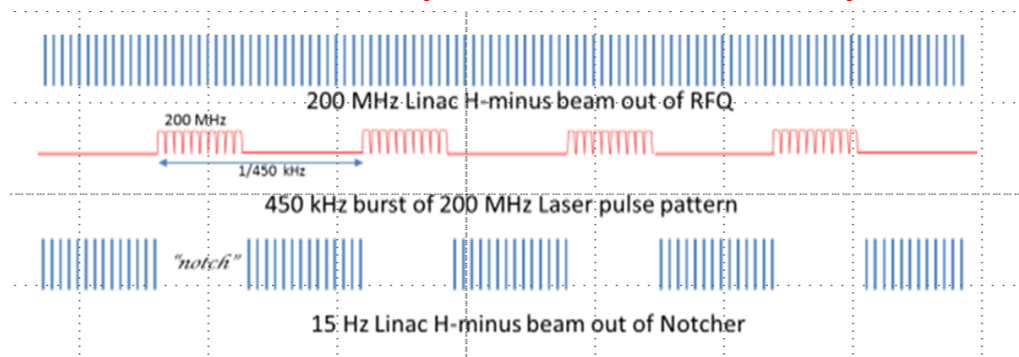
$$F_1 = N/N_0 = (1 - e^{-f_{CM} * \sigma(E) * \tau_{\text{crossing}}}) \quad \text{➤ 1 micron laser- near peak of cross section}$$

✓ Want to make process efficient -> minimize required laser pulse energy/average power



➤ Zig-zag optical cavity

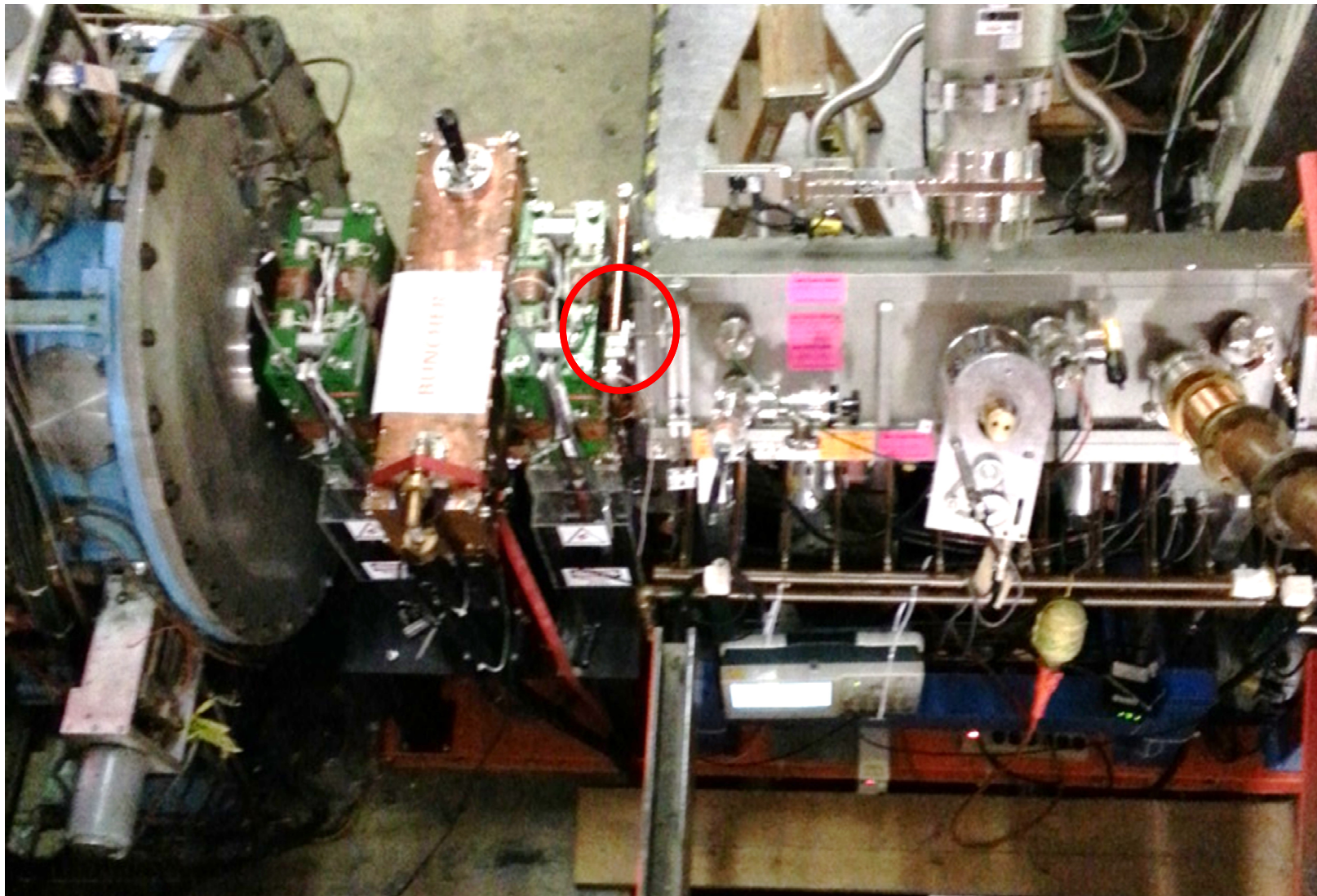
➤ Match laser pulses to bunch parameters



PIP Charge

- Develop a flexible laser system / optical cavity with the following attributes:
 - Create:
 - ✓ An arbitrary number of notches within a linac pulse
 - ✓ An arbitrary notch length for each notch
 - Provide:
 - ✓ 98% neutralization of the H- ions
 - Fit into the highly constrained space in the 750 keV MEBT where
 - ✓ component activation does not occur!
 - Create a system that is operationally robust
- This project is a task under the umbrella of PIP.

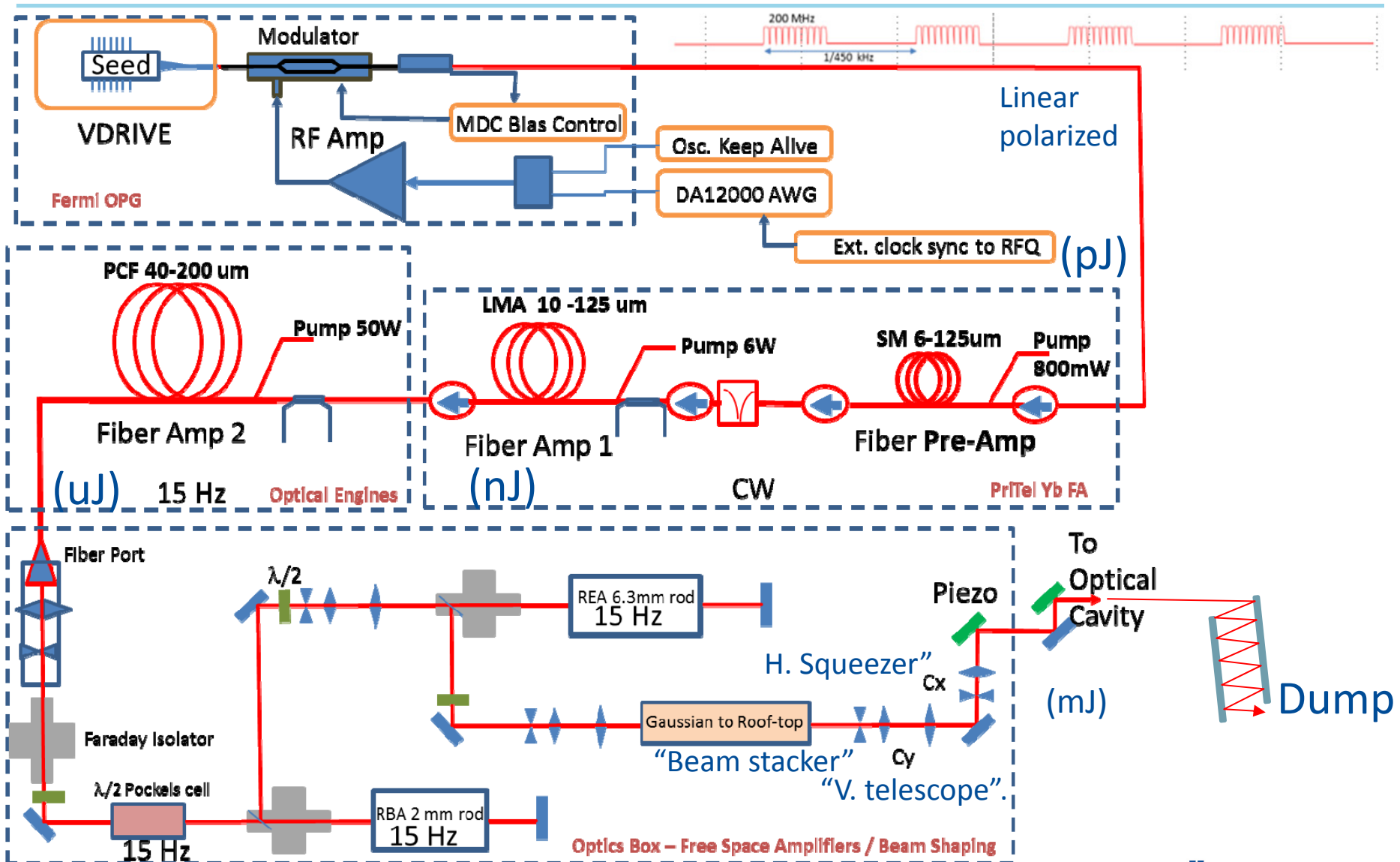
Location of the Zig-Zag Optical Cavity



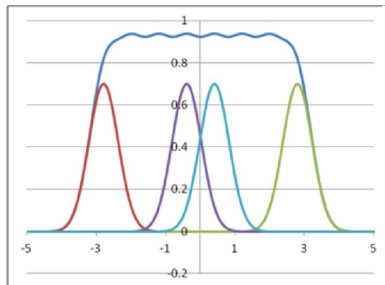
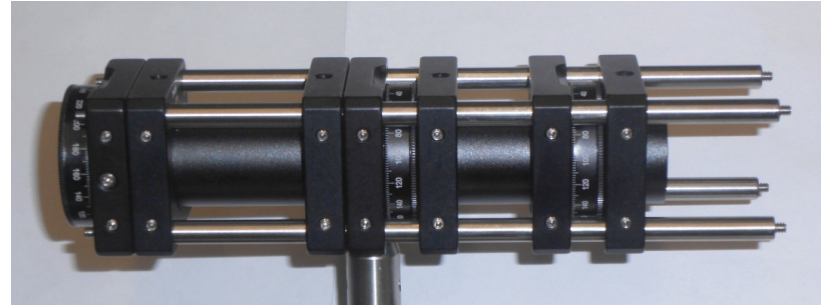
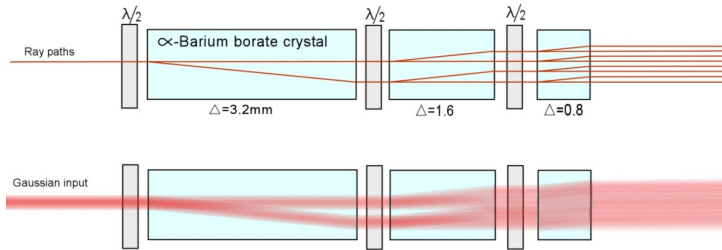
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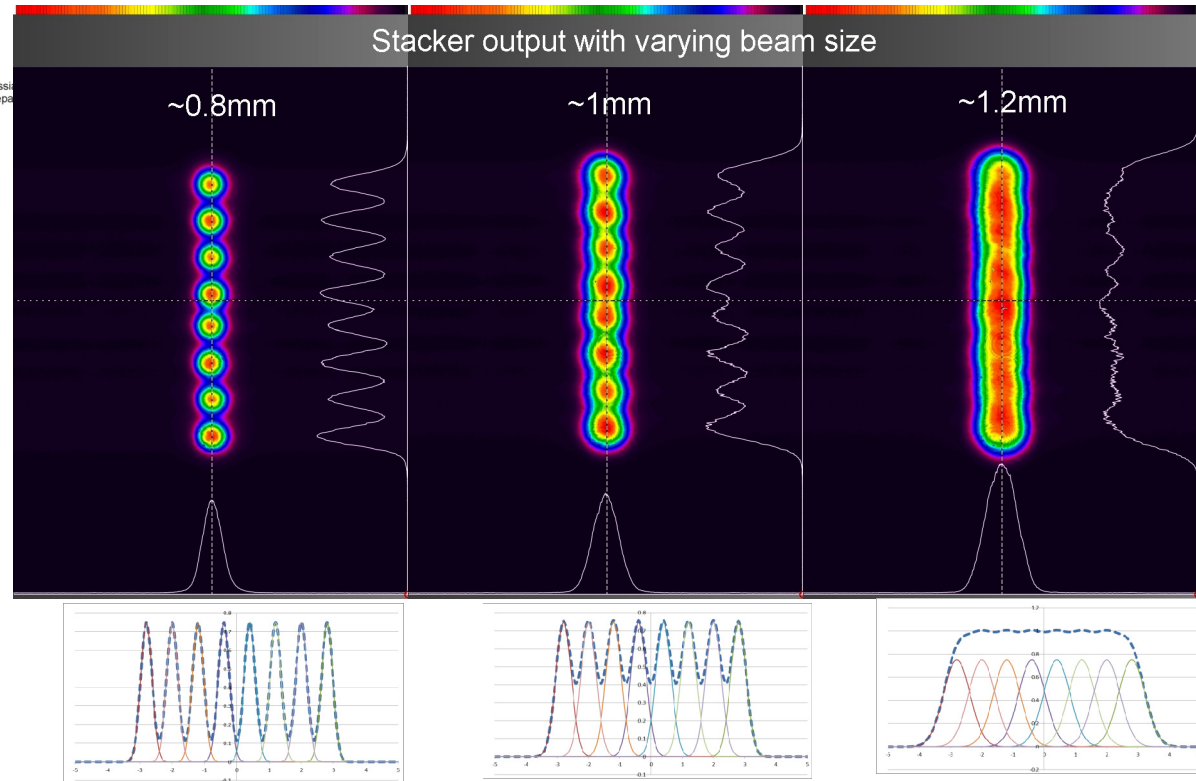
Laser System (Basic layout)



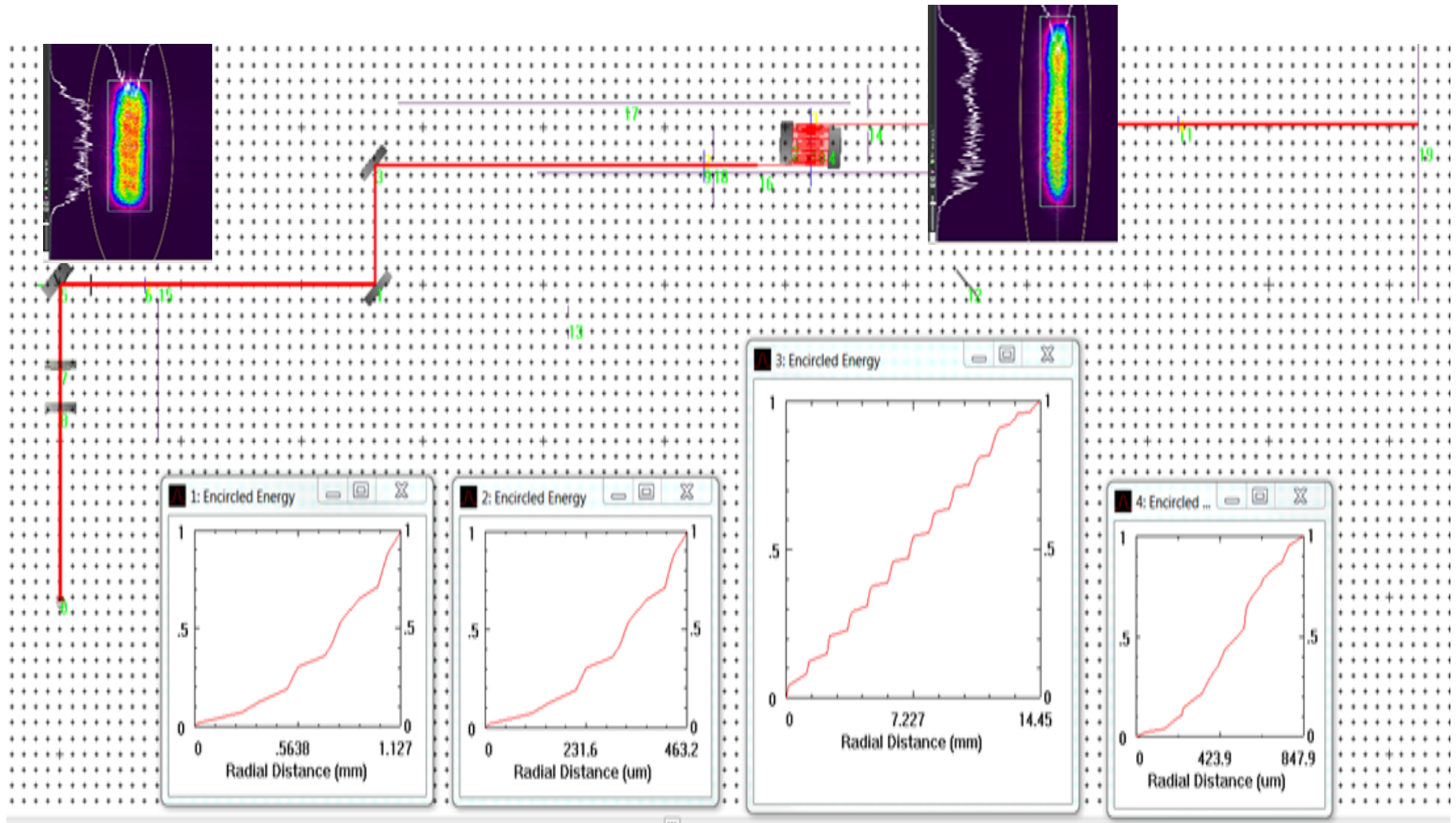
Creating a roof-top uniform profile



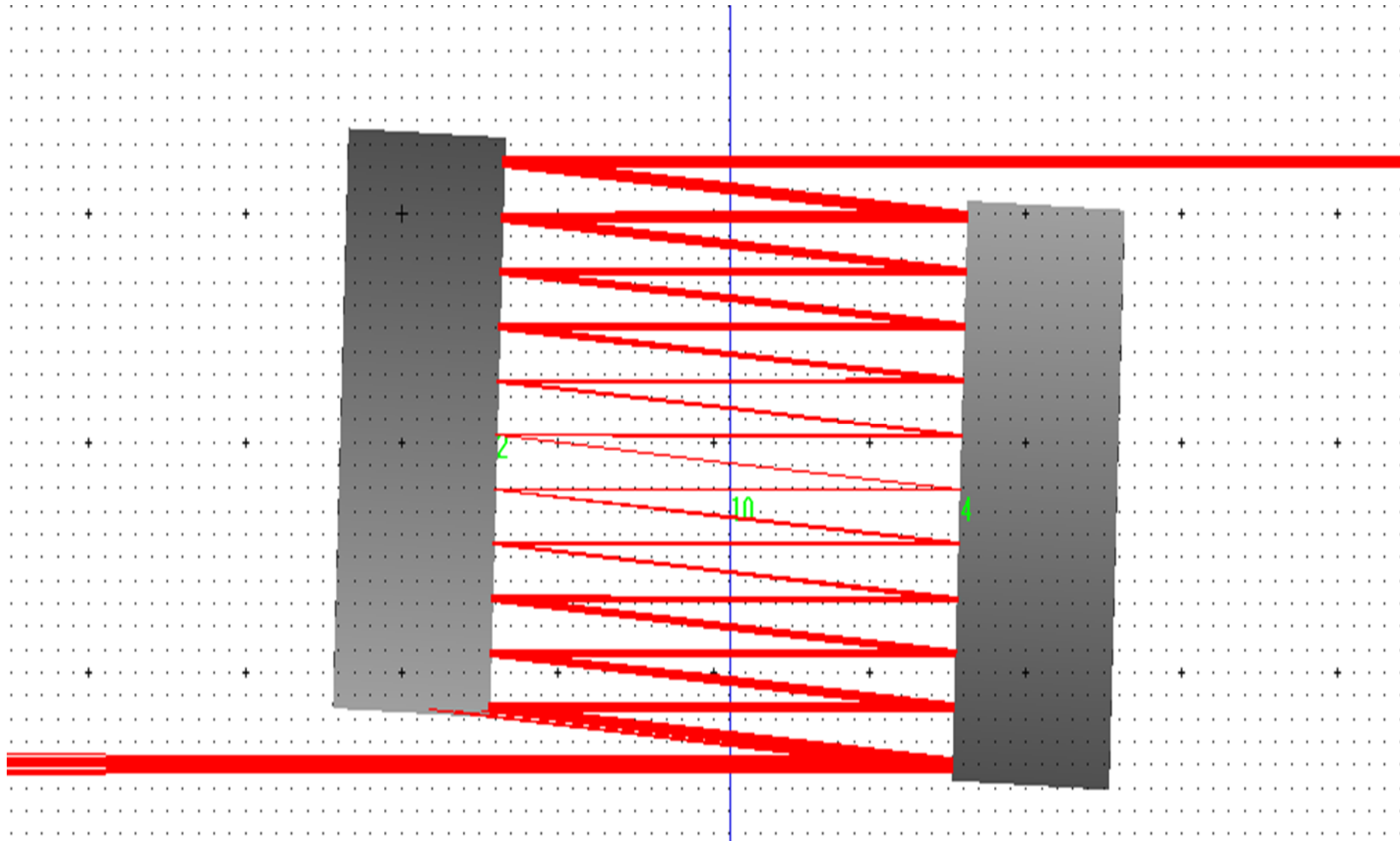
Simulation of 8 stacked gaussians
0.6mm sigma and 0.8mm separation
(4 of 8 beamlets shown).



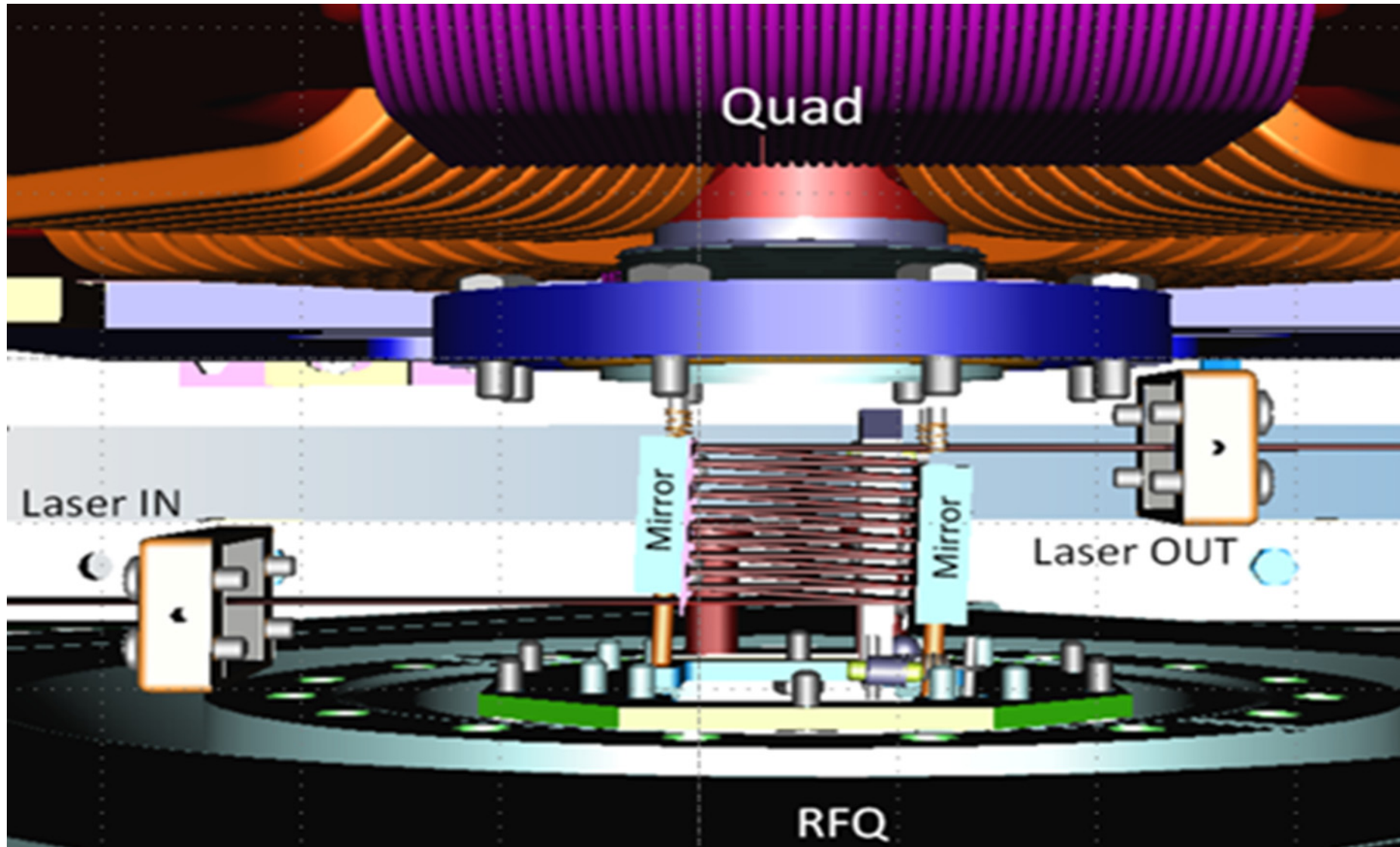
Optical Cavity



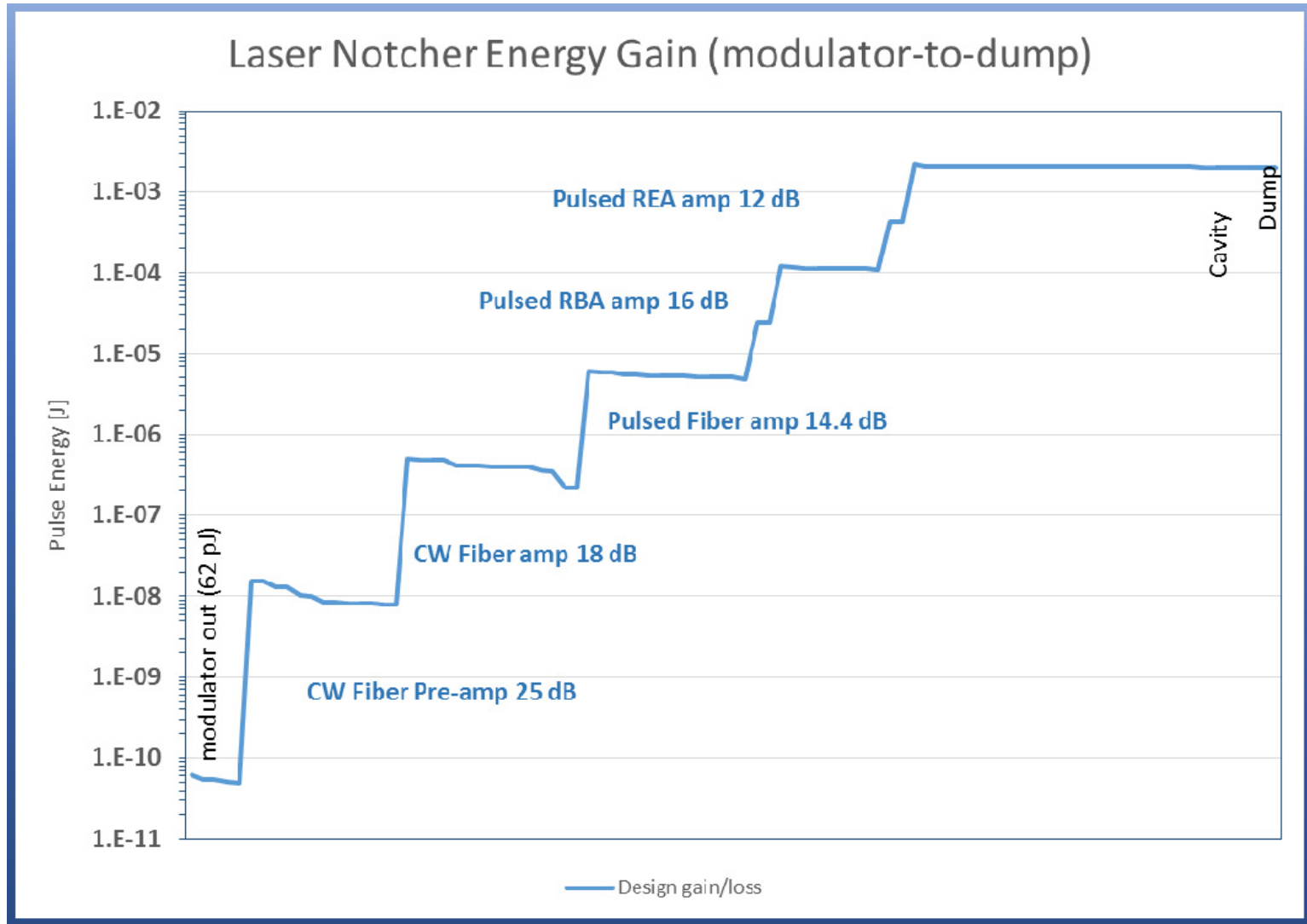
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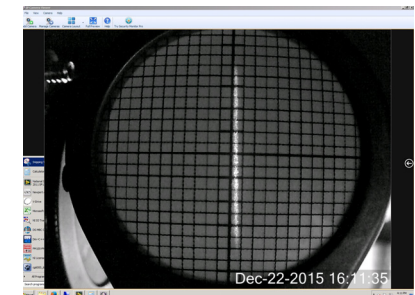
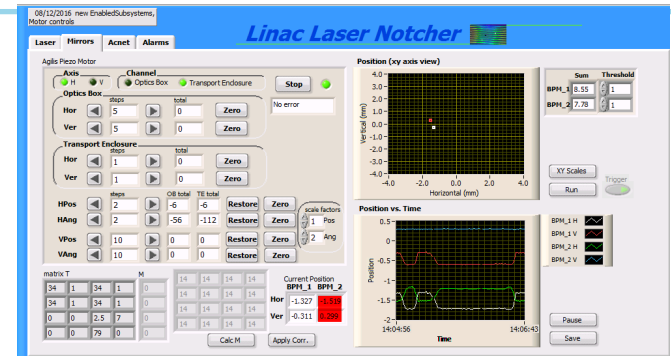


Pulse Energy Gain



Instrumentation

- Optical BPM's (fermi design)
 - Steering into cavity
- Fast photodiodes (2.1 GHz BW)
 - to monitor free space pulse peak
- Gated integrating spheres to measure power at 15 Hz in free space optics (fermi design)
- Pyroelectric energy meter in the Dump to measure the total laser energy each 15 Hz linac cycle.
- Video camera for
 - Qualitative profile measurement
 - monitoring laser position



Control Program

LLNMain.vi

File Edit View Project Operate Tools Window Help

09/08/2016 hide/show marker option

Linac Laser Notcher

Laser **Mirrors** **Acnet** **Alarms**

Grumman RBA

Cmd Current: 130 A
Sense Current: 130 A
Current Limit: 137 A
Pulse Width: 200.1 μ S
P.S. Voltage: 26.7 V
Duty Cycle: 0.4 %

Fault: Off
Flow: Off
E Stop: Off

Emission Reset Fault

Grumman REA

Cmd Current: 130 A
Sense Current: 0 A
Current Limit: 136.9 A
Pulse Width: 199.7 μ S
P.S. Voltage: 239.8 V
Duty Cycle: 0.4 %

Fault: Off
Flow: Off
E Stop: Off

Emission Reset Fault

PolyScience Chiller

Status: Run
Temp Set Point: 28 C
Set Point Read: 28.0 C
Read Temp: 28.0 C
Read Press: 22 psi
Read Flow: 3.1 gpm
Fault: System OK

ON Cmd Echo

V-Drive LD Controller

Set Temp: 8 $^{\circ}$ C
Temp: 8 $^{\circ}$ C
TEC Current: 0.45 mA
Set Current: 386 mA
Current: 386.07 mA
Optical Power: 156.17 mW

LD ON

Photline MBC DG

Bias Mode: AUTO
DC Bias Volts: 0 4.64 V
Transfer Lvl: 0 0 %
Dither Amp: 30 50 mV
Dither Freq: 1040 1000 Hz
PD Gain: 3 3
Mod Monitor: -4.77 V

MBC ON RESET

RBA Lambda 40-19

OVP: 32.00 V
UVL: 0.00 V
Voltage Setpoint: 29.00 V
Voltage Out: 29.00 V
Current Out: 0.41 A

Output ON Advanced

REA Lambda 300-5

OVP: 300.0 V
UVL: 0.00 V
Voltage Setpoint: 240.0 V
Voltage Out: 240.0 V
Current Out: 0.00 A

Output ON Advanced

Pritel YbFA

Control Status: Remote
Pump Status:
Auto Shut Down:
PreAmp Current: 650 mA
PwrAmp Input: 63 mW
PwrAmp Current: 7.46 A
Pump Current: 7.5 A

PUMP ON

Optical Engines Power Amp

Pulse Current: 5 5 A
Pulse Width: 45 44.7 μ S
Buck Voltage: 31 31 V

Advanced

Enables

- ☒ LD Permit
- ☒ LD Cntrl
- ☒ MBC DG
- ☒ YbFA
- ☒ Pwr Amp
- ☒ RBA
- ☒ REA
- ☒ Piezo Mtrs
- ☒ DA12000
- ☒ Chiller
- ☒ RBA Pwr
- ☒ REA Pwr

Com Status

Alarm

System

Shutdown Configuration

Save Settings Alarm Setup

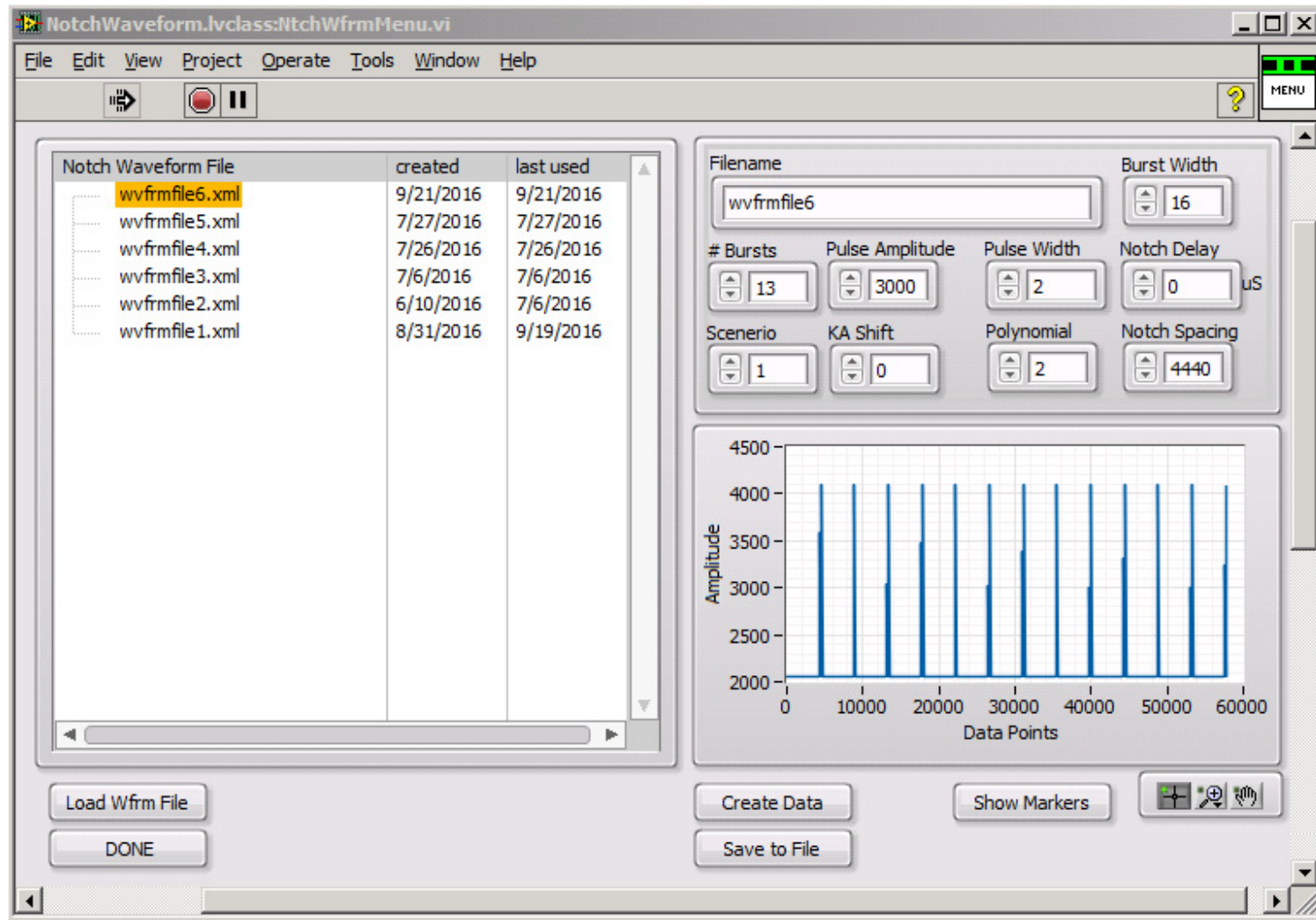
Control

Notch Wfrm KA ON RF AMP ON Align Mode OFF HP Mode OFF

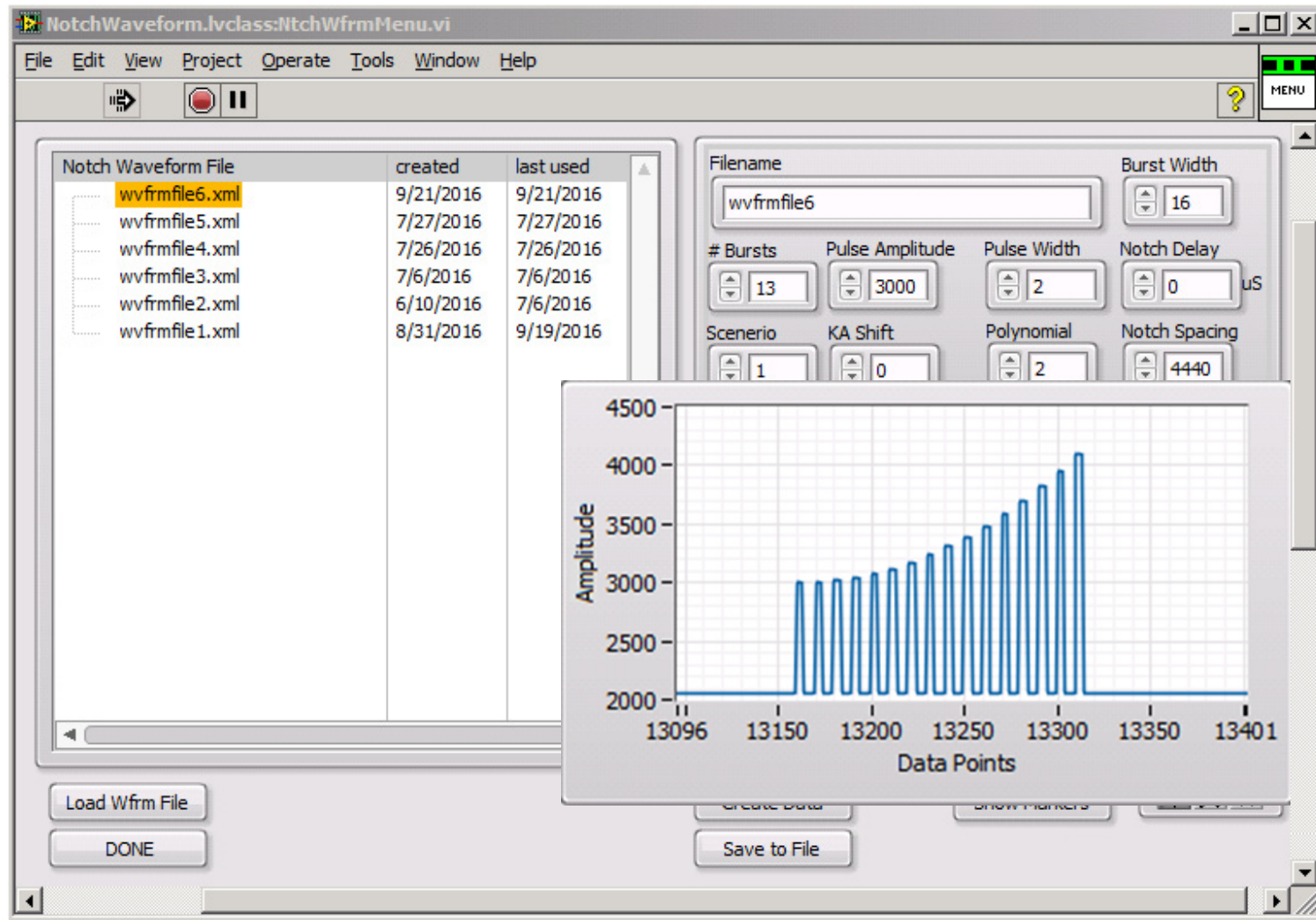
Status

09/24/2016 05:18:14 PM Opt Eng Set Pulse Current (A) 5.000000
09/24/2016 05:18:34 PM Opt Eng Set Pulse Current (A) 5.000000
09/24/2016 05:19:03 PM Grumman RBA Command Current (A) 60.00
09/24/2016 05:19:06 PM Grumman RBA Emission ON
09/24/2016 05:21:37 PM Grumman RBA Command Current (A) 130.00

Arbitrary Waveform Generation



Arbitrary Waveform Generation



July 2016 Commissioning Period (2weeks)

July Commissioning - Installation



Surveying the optical box



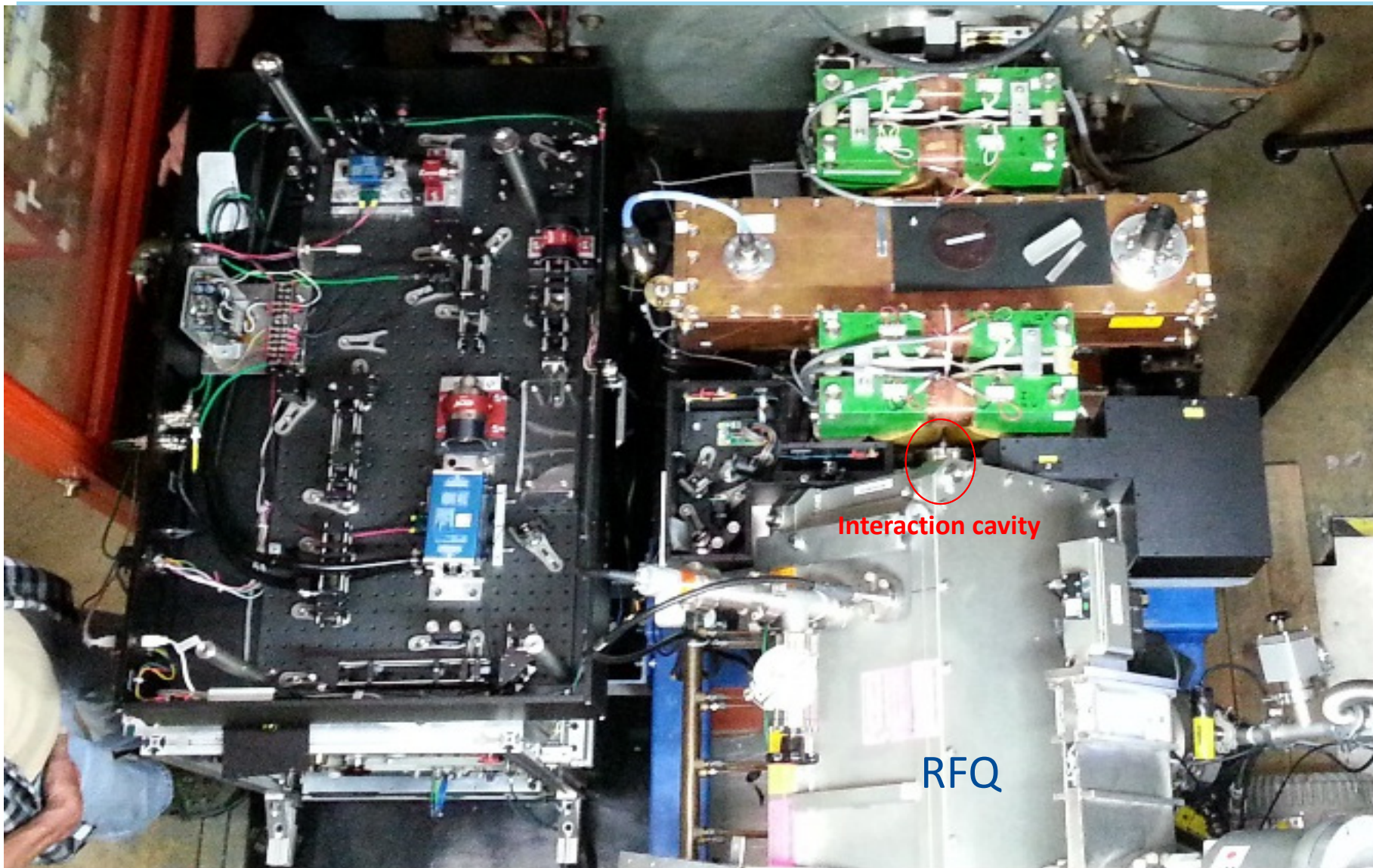
Electronics rack



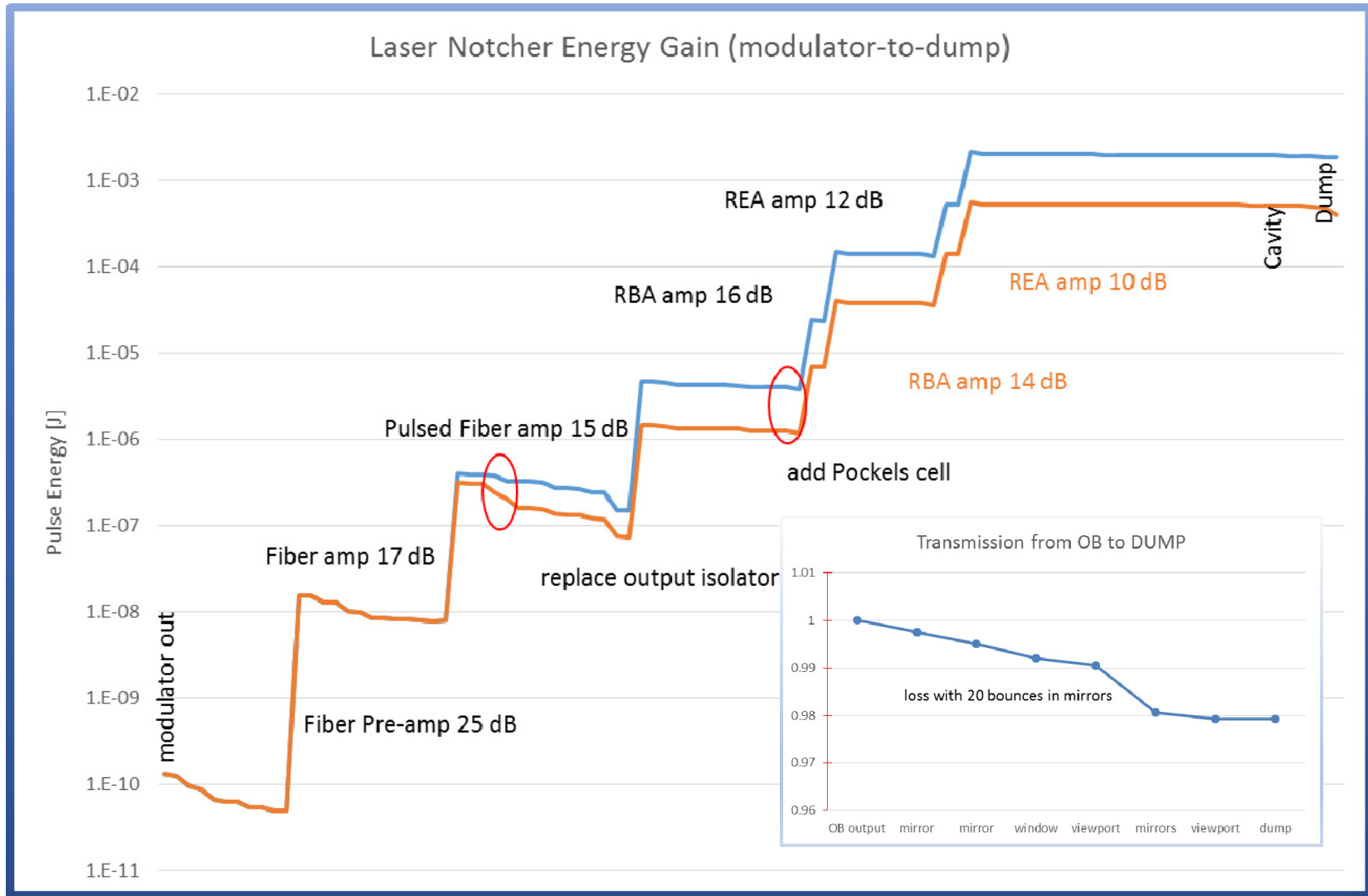
Pulsed fiber amplifier

- Tight quarters
- Light tight optical enclosures

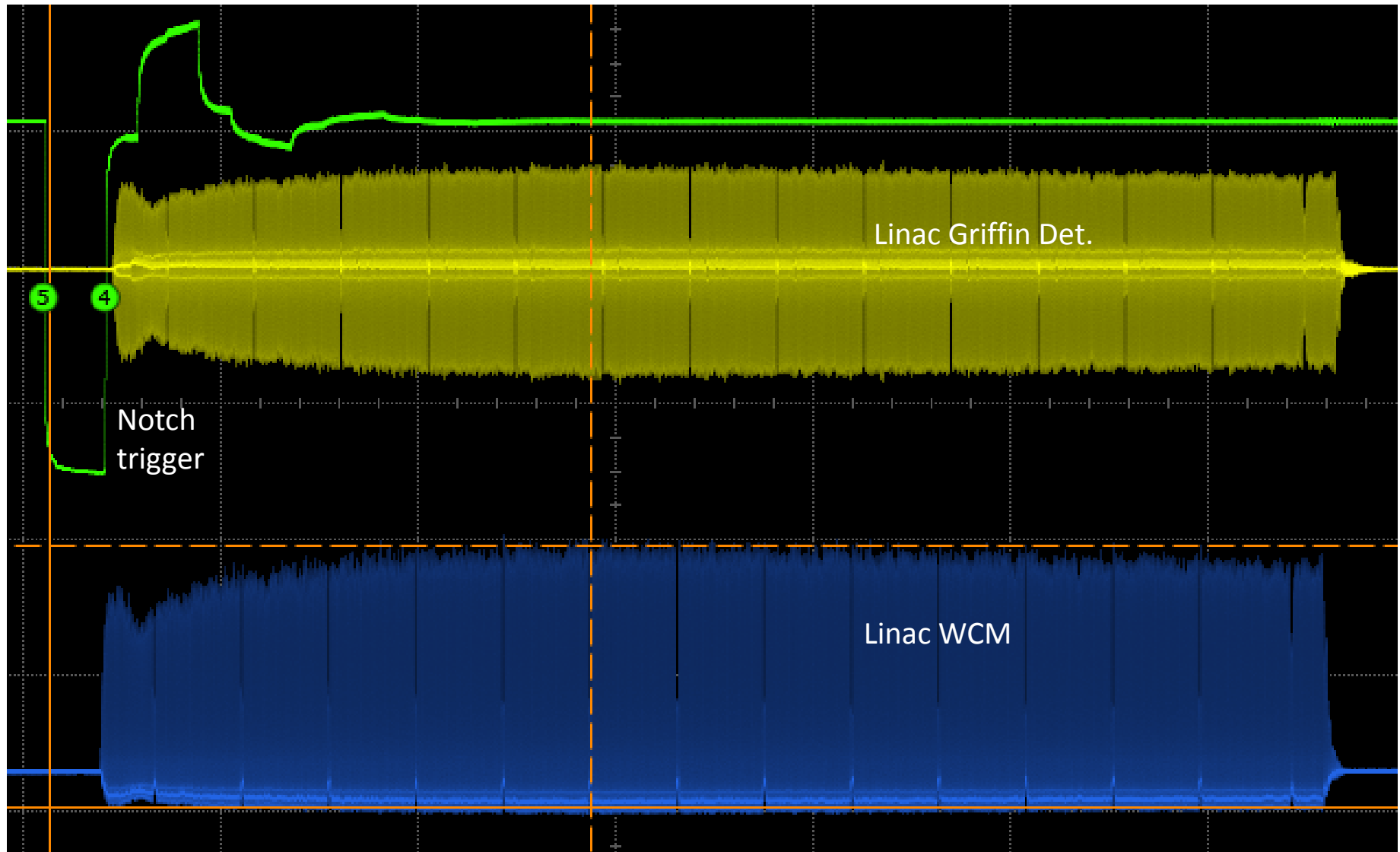
July Commissioning - Installation



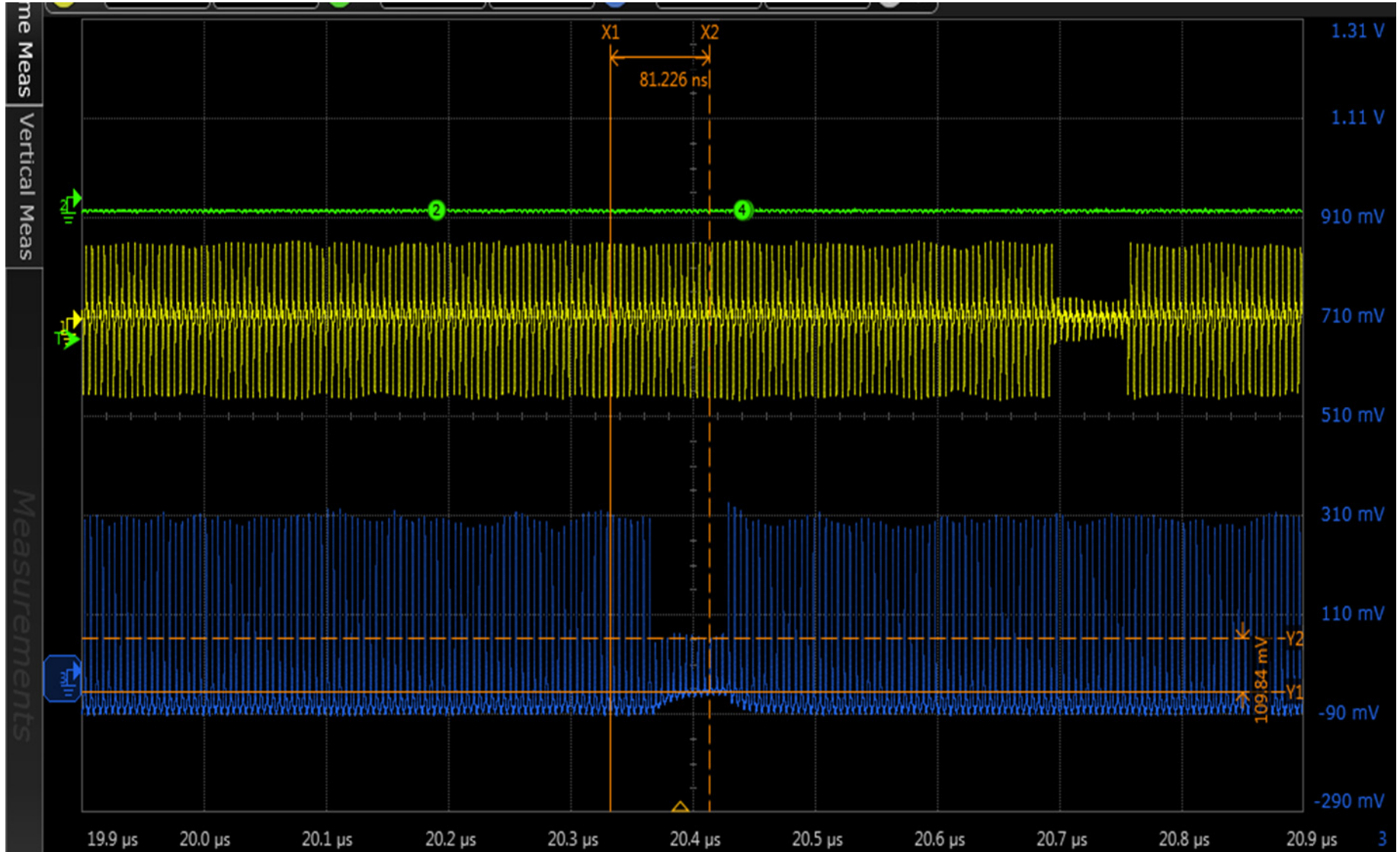
Pulse Energy Evolution through Laser Notcher System



First Observation of Multiple Notches in Linac Pulse



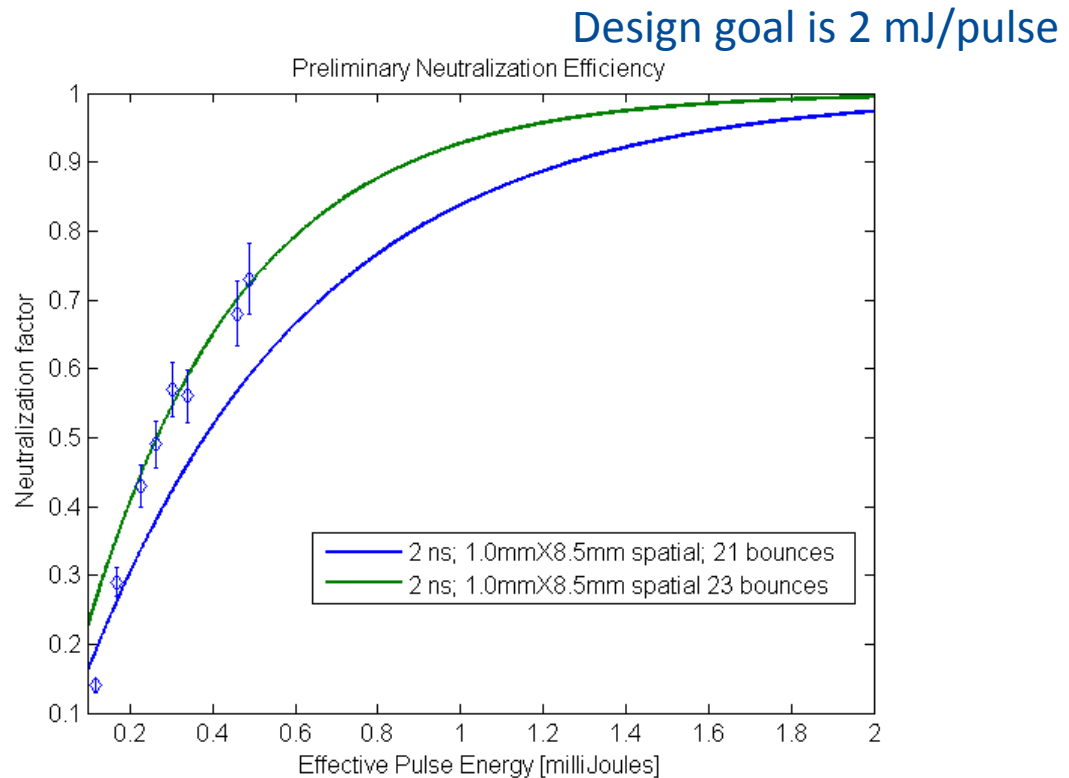
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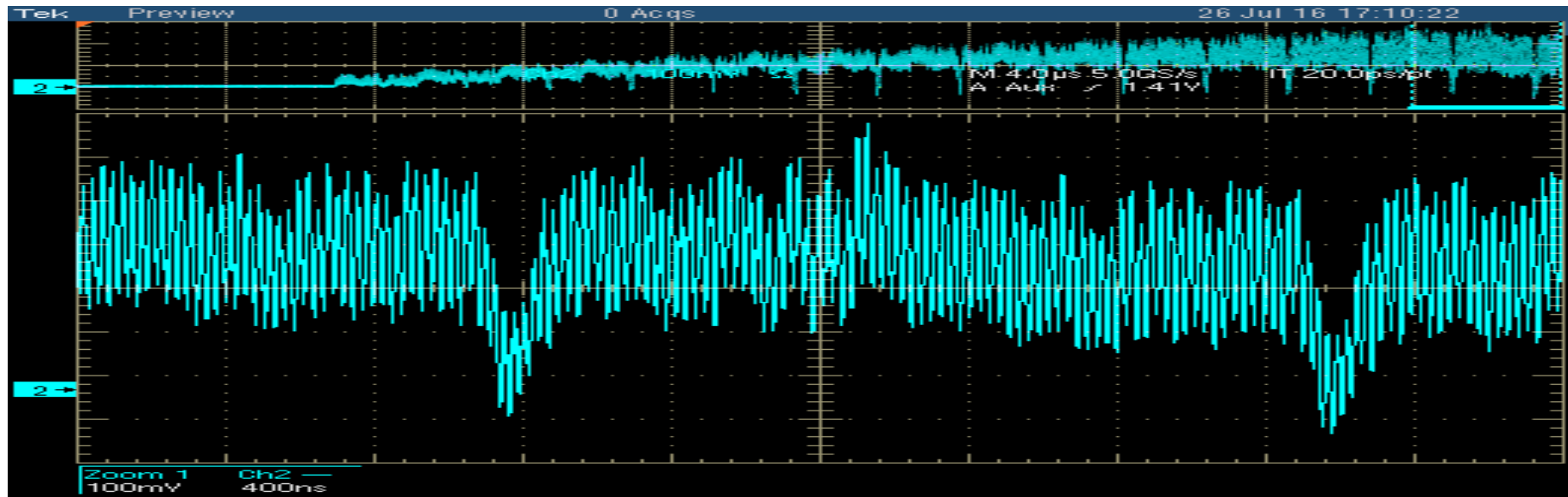
Preliminary Neutralization Efficiency

Vary laser pulse energy
Compare with estimation

Total Energy [mJ]	Energy in Notch Pulse [mJ]	200MHz pulse energy [mJ]	Measured Neutralization
47	18.471	0.118	0.14
66	25.938	0.166	0.29
90	35.37	0.227	0.43
104	40.872	0.262	0.49
120	47.16	0.302	0.57
135	53.055	0.340	0.56
184	72.312	0.464	0.68
198	77.814	0.499	0.73

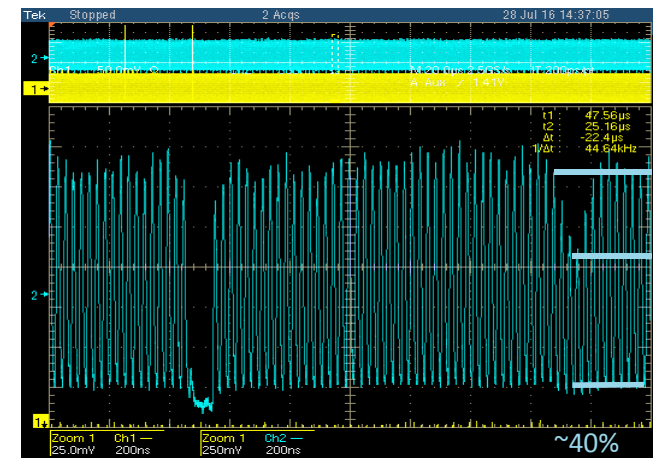


Notches Observed in Booster



- a) Upper: Booster phase detector during injection showing notch formation from linac
- b) Lower: Notch in the 14th and 15th turn showing a bottom width with 38 MHz on (width at bottom ~ 60 ns)
- c) Bottom Right: A few ms into cycle showing notch created in Booster (left) and notch created in linac (right)

➤ Simulations of capture details give tuning guidance to keep notch clean to a few % level



Conclusions/Take Away

- ✓ Successfully demonstrated “for the first time anywhere” the neutralization of multiple sections of H- ion bunch structure out of the RFQ.
- ✓ Although the laser system was not at design intensity (0.5 mJ/pulse) we demonstrated ~70% neutralization in the Linac pulse train.
- ✓ Although we saw survival of the notch ms after injection, it is clear that we need to address the survival of the notch in Booster when we turn back on (simulations encouraging)
and
- ✓ This technique may be used for bunch-by-bunch manipulations of H- ions for other applications...

Thank you for your attention